

Project Euler

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Problem 2: Even Fibonacci Numbers

Even Fibonacci Numbers

Problem 2



Each new term in the Fibonacci sequence is generated by adding the previous two terms. By starting with 1 and 2, the first 10 terms will be:

1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.

Answer:

Check

$$2, 8, 34, 144 \dots$$

These terms follow a different pattern from the regular Fibonacci sequence that follows

$$f(n+1) = f(n) + f(n-1)$$

The even term instead follows the recurrence as follows:

$$f(n+1) = 4f(n) + f(n-1)$$

$$f(1) = 2, \quad f(2) = 8,$$

$$f(3) = 4f(2) + f(1) = 4(8) + 2 = 34,$$

$$f(4) = 4f(3) + f(2) = 4(34) + 8 = 144,$$

$$f(5) = 4f(4) + f(3) = 4(144) + 34 = 610,$$

$$f(6) = 4f(5) + f(4) = 4(610) + 144 = 2584,$$

$$f(7) = 4f(6) + f(5) = 4(2584) + 610 = 10946,$$

$$f(8) = 4f(7) + f(6) = 4(10946) + 2584 = 46368,$$

$$f(9) = 4f(8) + f(7) = 4(46368) + 10946 = 196418,$$

$$f(10) = 4f(9) + f(8) = 4(196418) + 46368 = 832040,$$

$$f(11) = 4f(10) + f(9) = 4(832040) + 196418 = 3524578,$$

$$f(12) = 4f(11) + f(10) = 4(3524578) + 832040 = 14930352.$$

$$f(12) > 4,000,000, \text{ hence stop at } f(11)$$

$$S = 2 + 8 + 34 + 144 + 610 + 2584 + 10946 + 46368 + 196418 + 832040 + 3524578$$

$$s = 4,613,732$$



Congratulations, the answer you gave to problem 2 is correct.